Practice workbook · Define specialised cells, tissues and organs and identify human tissues

Definitions of specialised cell, tissue and organ, naming epithelial and muscle tissues, comparing cheek and tracheal cells, and describing cardiac, smooth and skeletal muscle.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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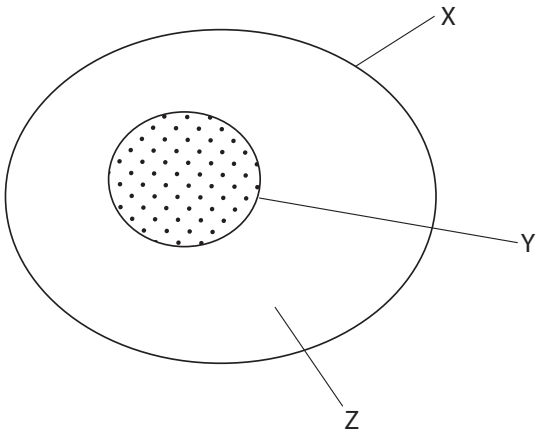
Type ID: HBI-CELLS-TISSUES-ORGANS

# 1. Question 1 · part a.iii · 2 marks

June 2022 Paper 1R

Worked-example retry · Attempt part a.iii only.

**1** (a) The diagram shows a human body cell.




(i) Name the parts X, Y and Z.

**(3)**

X .....

Y .....

Z .....

(ii) Name one structure in the cell that is not shown in the diagram.

**(1)**

.....

(iii) Explain how body cells are organised to form organs.

**(2)**

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# 1. Question 1 · part a.iii · 2 marks

June 2022 Paper 1R

Worked-example retry · Attempt part a.iii only.

(b) The boxes list three processes that transport substances in and out of cells.

Complete the diagram by drawing one straight line from each process to its correct definition.

**(3)**

| Process | Definition |
|---|---|
| | requires energy to move molecules against a concentration gradient |
| osmosis | |
| | water moves from a high water potential to a low water potential |
| diffusion | |
| | water moves from a low water potential to a high water potential |
| active transport | |
| | molecules move down a concentration gradient |

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## 1. Question 1 · part a.iii · 2 marks · Working and answer

Working and answer

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# 2. Question 2 · part b.i · 3 marks

November 2023 Paper 2

Further practice · Attempt part b.i only.

**2** (a) Draw a labelled diagram of a cheek cell from inside a mouth as seen under a light microscope.

**(4)**

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(b) (i) Explain what is meant by the term **tissue**.

**(3)**

(ii) State the type of tissue that contains cheek cells.

**(1)**

(iii) Name two other types of tissue.

**(2)**

1 .....

2 .....

**(Total for Question 2 = 10 marks)**

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## 2. Question 2 · part b.i · 3 marks · Working and answer

Working and answer

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# 3. Question 3 · part a · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a only.

**3** (a) Describe what is meant by a specialised cell.

**(2)**

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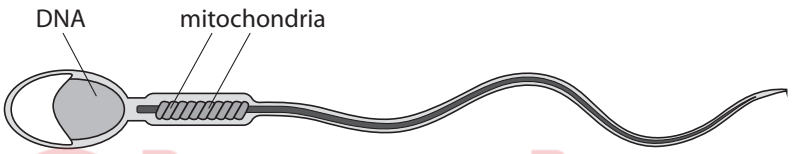
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(b) The sperm cell is an example of a specialised cell.

The diagram shows a sperm cell.

NOT TO SCALE




(i) The sperm cell in the diagram has an actual length of 0.0060 mm.

The length of the image of the sperm cell is 84 mm.

Calculate the magnification of the sperm cell.

**(2)**

magnification = × .....

(ii) The head of the sperm contains DNA.

Give the difference in the amount of DNA found in the head of the sperm and in a nucleus of a body cell.

**(1)**

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## 3. Question 3 · part a · 2 marks · Working and answer

Working and answer

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# 4. Question 7 · part b.i · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b.i only.

**7** (a) The nervous system sends electrical impulses to parts of the body in response to an external stimulus.

(i) Name the structure in the nervous system that detects an external stimulus.

**(1)**

(ii) Name the structure that transmits electrical impulses directly to the central nervous system.

**(1)**

(b) Neurones are specialised cells.

(i) Describe what is meant by a specialised cell.

**(2)**

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## 4. Question 7 · part b.i · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b.i only.

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(ii) A motor neurone is a specialised cell.

Draw a diagram of a motor neurone.

Your diagram should include these labels.

- nucleus
- cell body
- myelin sheath

**(4)**

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(iii) Describe the function of a motor neurone in the nervous system.

**(2)**

## 4. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

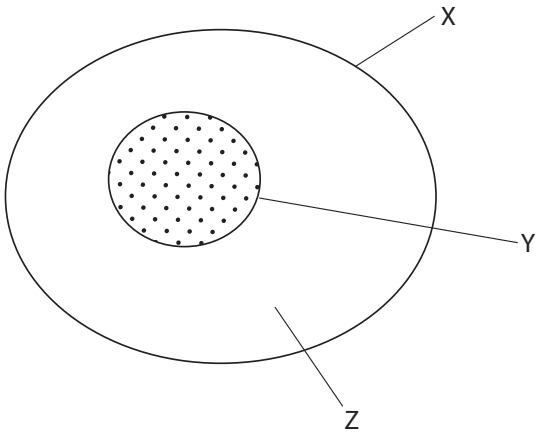
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# 5. Question 1 · part c.i · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.i only.

**1** (a) The diagram shows a human body cell.




(i) Name the parts X, Y and Z.

**(3)**

X .....

Y .....

Z .....

(ii) Name one structure in the cell that is not shown in the diagram.

**(1)**

.....

(iii) Explain how body cells are organised to form organs.

**(2)**

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# 5. Question 1 · part c.i · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.i only.

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(b) The boxes list three processes that transport substances in and out of cells.

Complete the diagram by drawing one straight line from each process to its correct definition.

**(3)**

| Process | Definition |
|---|---|
| | requires energy to move molecules against a concentration gradient |
| osmosis | |
| | water moves from a high water potential to a low water potential |
| diffusion | |
| | water moves from a low water potential to a high water potential |
| active transport | |
| | molecules move down a concentration gradient |

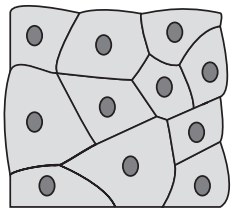
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# 5. Question 1 · part c.i · 2 marks

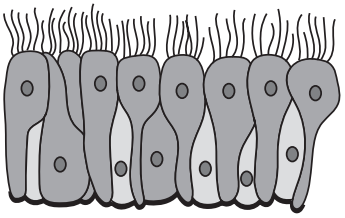
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Further practice · Attempt part c.i only.

(c) (i) The diagram shows two types of epithelial tissue, A and B.



Tissue A



Tissue B

(Source: © Timonina/Shutterstock)

Give the name of each type of epithelial tissue.

**(2)**

A .....

B .....

(ii) An epithelial cell has an actual width of 60 micrometres (μm).

The cell is magnified 40 times.

Calculate the image width of this cell in mm.

[1 mm = 1000 μm]

**(3)**

image width = ..... mm

**(Total for Question 1 = 14 marks)**

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## 5. Question 1 · part c.i · 2 marks · Working and answer

Working and answer

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# 6. Question 6 · part c · 3 marks

June 2023 Paper 1R

Further practice · Attempt part c only.

**6** The lining of the inside of the cheek is made of squamous epithelial cells.

(a) Draw a labelled diagram to show a cheek cell when seen using a light microscope.

**(4)**

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(b) Describe how a sample of cheek cells would be prepared to view using a light microscope.

**(3)**

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## 6. Question 6 · part c · 3 marks

June 2023 Paper 1R

Further practice · Attempt part c only.

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(c) State three differences between a cheek cell and a cell from the lining of the trachea.

**(3)**

1 .....

.....

2 .....

.....

3 .....

.....

**(Total for Question 6 = 10 marks)**

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Working and answer

# 7. Question 2 · part b.ii · 1 mark

November 2023 Paper 2

Further practice · Attempt part b.ii only.

**2** (a) Draw a labelled diagram of a cheek cell from inside a mouth as seen under a light microscope.

**(4)**

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(b) (i) Explain what is meant by the term **tissue**.

**(3)**

(ii) State the type of tissue that contains cheek cells.

**(1)**

(iii) Name two other types of tissue.

**(2)**

1 .....

2 .....

**(Total for Question 2 = 10 marks)**

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## 7. Question 2 · part b.ii · 1 mark · Working and answer

Working and answer

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## 8. Question 2 · part b.iii · 2 marks

November 2023 Paper 2

Further practice · Attempt part b.iii only.

**2** (a) Draw a labelled diagram of a cheek cell from inside a mouth as seen under a light microscope.

**(4)**

(b) (i) Explain what is meant by the term **tissue**.

**(3)**

(ii) State the type of tissue that contains cheek cells.

**(1)**

(iii) Name two other types of tissue.

**(2)**

1 .....

2 .....

**(Total for Question 2 = 10 marks)**

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## 8. Question 2 · part b.iii · 2 marks · Working and answer

Working and answer

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# 9. Question 5 · part a · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a only.

**5** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The heart is a double-pump that is part of the human circulatory system. It helps ensure that the blood and its components reach all cells in the body.

Some components of the blood are made in the bone marrow. An average adult with a mass of 70 kg has 5.5 $dm^3$ of blood. About 45% of the blood is made up of different cells. There are several different types of cell in the blood but red blood cells are the most common. In males, 1.0 $dm^3$ of blood contains $5.9 \times 10^{12}$ red blood cells. In females the same volume of blood contains $5.5 \times 10^{12}$ red blood cells. Haemoglobin makes up a large part of red blood cells. This protein is vital to the function of red blood cells.

Red blood cells are transported around the body in a liquid known as plasma. Only 10% of blood plasma is made up of components that are used or produced by body cells. The rest of the plasma is made up of water. Some components of blood plasma are used by body cells in chemical reactions, some act as messengers or have a role in blood clotting and others are excreted from the body as waste.

(a) Name the main tissue that makes up the walls of the heart. **(1)**

(b) Explain why the heart is known as a double-pump. (Lines 1 to 2) **(3)**

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## 9. Question 5 · part a · 1 mark · Working and answer

Working and answer

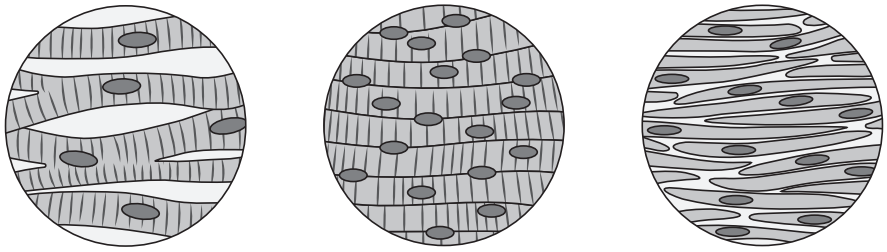
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## 10. Question 6 · part a.i · 6 marks

November 2024 Paper 1

Further practice · Attempt part a.i only.

**6** (a) The diagram shows three types of muscle tissue.



cardiac muscle | voluntary muscle | involuntary muscle

(Source: © udaix/Shutterstock)

(i) Describe the structure of each type of muscle tissue.

**(6)**

cardiac

voluntary

involuntary

(ii) Give one structure of the human body where involuntary muscle tissue is found.

**(1)**

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## 10. Question 6 · part a.i · 6 marks · Working and answer

Working and answer

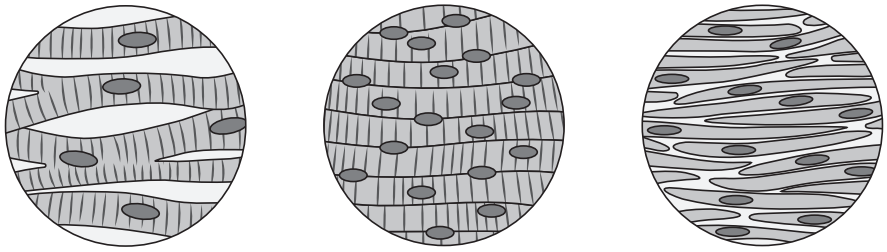
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## 11. Question 6 · part a.ii · 1 mark

November 2024 Paper 1

Further practice · Attempt part a.ii only.

**6** (a) The diagram shows three types of muscle tissue.



cardiac muscle | voluntary muscle | involuntary muscle

(Source: © udaix/Shutterstock)

(i) Describe the structure of each type of muscle tissue.

**(6)**

cardiac

voluntary

involuntary

(ii) Give one structure of the human body where involuntary muscle tissue is found.

**(1)**

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# 11. Question 6 · part a.ii · 1 mark

November 2024 Paper 1

Further practice · Attempt part a.ii only.

(b) Magnetic resonance imaging (MRI) can be used to measure the percentage muscle mass of an individual.

In one year, 20 female athletes had their percentage muscle mass measured using MRI.

These are the results.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 74% | 75% | 70% | 82% | 74% | 74% | 78% | 82% | 70% | 82% |
| 78% | 74% | 82% | 78% | 82% | 75% | 84% | 72% | 82% | 72% |

(i) Calculate the mean percentage muscle mass.

**(2)**

mean percentage muscle mass = .....%

(ii) Draw a frequency table to show the number of each percentage muscle mass recorded for the female athletes.

**(3)**

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(iii) Determine the mode of these percentage muscle masses.

**(1)**

mode = .....

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## 11. Question 6 · part a.ii · 1 mark · Working and answer

Working and answer

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